

UFO — Integrated Single-Seat Vehicle: a 4-Domain Capstone of Verify-Native Propulsion

RTSC · FUSION · ANTIMATTER · CERN — a 7-Stage Propulsion Ladder with a Sharp Proven/UNPROVEN Wall, a 4-Domain Integration Map, and the Full demiurge 7-Verb Pipeline (`spec`→`handoff`)

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May 26, 2026

Abstract

We present **UFO**, the *capstone* (integration) volume of the demiurge stack: a single-seat vehicle that ties together four independently-verified sibling domains — **RTSC** (real-temperature superconducting magnets), **FUSION** (magnetic-confinement / MHD propulsion), **ANTIMATTER** (γ -rocket annihilation propulsion), and **CERN** (accelerator field / particle infrastructure) — into one carrier and carries it through the full demiurge seven-verb pipeline (`spec`→`structure`→`design`→`analyze` $\circ$ →`synthesize`→`verify`→`handoff`), non-wet-lab, end to end. The physical spine is a **seven-stage propulsion ladder** (§3): Stage 1 hover (Meissner levitation on an RTSC 48 T solenoid), Stage 2 cruise (MHD+fusion body-force thrust), Stage 3 orbital (γ -rocket annihilation exhaust), then Stage 4 warp (Alcubierre), Stage 5 wormhole (Morris–Thorne), Stage 6 dim-jump (Kaluza–Klein ladder), and Stage 7 dim-use (a composite $\tau=4$ cycle). The ladder carries a deliberately *sharp honesty wall*: Stages 1–3 are closed-form verified (■, inherited verbatim from the sibling domains — `ioffe_loop_bz`, `triple_product`, `pair_threshold_total+rel_kinetic_from_p`, all SUPPORTED-NUMERICAL at $|\Delta|=0$, atlas hash 663698a0), while Stages 4–7 are a *13-falsifier preregister*, *ALL OPEN* ($\circ$, academically **UNPROVEN**). We do *not* claim warp, wormhole, dimensional jump, or dimensional use is achieved; we register, for each, the precise measurement that would falsify it. The capstone’s headline verdict is honest: **absorbed=false**. Five non-wet-lab digital-twin gates (CFD, EM 3D, structural FEA, the $\circ$ fixed-point coupling, and one cross-coupled gate) have not converged to a body-solve (they are deferred to pool/cloud heavy simulation), and Stages 4–7 are all open — so the integration bit cannot flip. A design that clears all non-wet-lab gates is a *witness of integrated-vehicle feasibility*, never a built or flown craft. Across the ladder we observe a shared lattice constant $B^* = \sigma \cdot \tau = 48$ that recurs in RTSC (H_{c2}), FUSION (Lawson target), and the UFO Meissner lift — but this is an *arithmetic-reuse observation*, not a proof of integrated physics. Bit-for-bit reproduction artifacts (verify ledger, PR roll, integration map, session journal) ship under `companion/`.

1 Introduction

The four sibling domains of the demiurge stack each answer a single, physically distinct propulsion question to a verified closed form: RTSC asks how strong a real-temperature superconducting magnet can be (and how much lift its Meissner fringe field provides); FUSION asks whether a magnetically-confined plasma can both ignite and exert an MHD body-force thrust; ANTIMATTER asks what specific impulse an annihilation γ -rocket can reach; and CERN asks how accelerator-grade fields and relativistic-kinematics infrastructure are produced and verified.

Each domain has its own monograph and its own verify ledger. None of them, on its own, answers the *integration* question: *can a single carrier consume all four, in sequence, as the rungs of one propulsion ladder?*

UFO is the capstone that asks exactly that. It is not a fifth physics domain — it adds essentially no new physics atoms. It is an *integration volume*: it consumes the four siblings’ verified records, places them on a seven-stage propulsion ladder, and runs the whole vehicle through the demiurge seven-verb design pipeline (**spec**→**handoff**) to a non-wet-lab completed form (§6). The contribution is therefore architectural honesty under integration pressure: keeping a sharp wall between the three rungs that are genuinely closed-form-verified and the four rungs that are open, unproven speculation — even as they sit in one table, on one vehicle, under one title.

Contributions.

1. A **seven-stage propulsion ladder** (§3) that places the four sibling domains in one physical path, with a *sharp proven/UNPROVEN wall* ($\parallel$) between Stages 1–3 (■, sibling-inherited closed-form) and Stages 4–7 (○, 13-falsifier preregister, ALL OPEN).
2. A **4-domain integration map** (§4, Table 2) keyed stage-by-stage to the sibling domain each rung consumes (hover←RTSC, cruise←FUSION, orbital←ANTIMATTER, fields/particles←CERN), with the shared $B^* = \sigma \cdot \tau = 48$ arithmetic-reuse observation flagged as *observed, not proven*.
3. The **full demiurge seven-verb pipeline** (§6) carried to a non-wet-lab completed form: every verb (**spec**, **structure**, **design**, **analyze** ○, **synthesize**, **verify**, **handoff**) has a landed artifact.
4. A **cross-domain reuse lattice** (§7) and the honest interpretation of its recurring lattice constants ($\sigma \cdot \tau = 48$, $\sigma \cdot \varphi = n \cdot \tau = 24$): arithmetic closure $\neq$ physical closure.
5. An honest **absorbed=false gate inventory** (§8): five non-wet-lab ○ gates pending body-solve plus the 13 open Stage-4–7 falsifiers, so the integration bit honestly stays **false**.
6. A clean **honesty boundary** (§9): we never claim the vehicle is build-ready, and we never present a ○ Stage-4–7 claim as proven. The wall is the contract.

A high-level view of the integrated vehicle and the proven/UNPROVEN wall is shown in Fig. 1.

Document structure. This is the *monograph* (capstone) form of UFO: the body (§§1–11) is the self-contained integration argument, and the lettered appendices (A–L) are the full data record — the three closed-form stage folds (A–C), the Stage-4–7 falsifier preregister (D), the five sub-axes (E), the seven-verb pipeline (F), the cross-domain integration lattice (G), the meta lattice/limit-breakthrough policy (H), the verify ledger (I), the **absorbed=false** gate inventory (J), the PR roll (K), and the reproducibility runbook (L). A reader can audit any body claim down to its sibling-domain atom by following the appendix it cites.

2 Background — the Capstone Thesis and the Verify Tier Rubric

The integration thesis. A magnetic-levitation magnet, a fusion-MHD thruster, and an antimatter γ -rocket each clear a different physical regime: hover near the ground, cruise through the atmosphere, and accelerate to orbital/escape velocity. The capstone thesis is that these are not three separate vehicles but three *rungs* of one ladder, mounted on one disc-shaped single-seat carrier, switched between by closed-form altitude/velocity/ Δv triggers. Stages 4–7 are notionally the upper rungs of the same ladder (interstellar and beyond), but — and this is

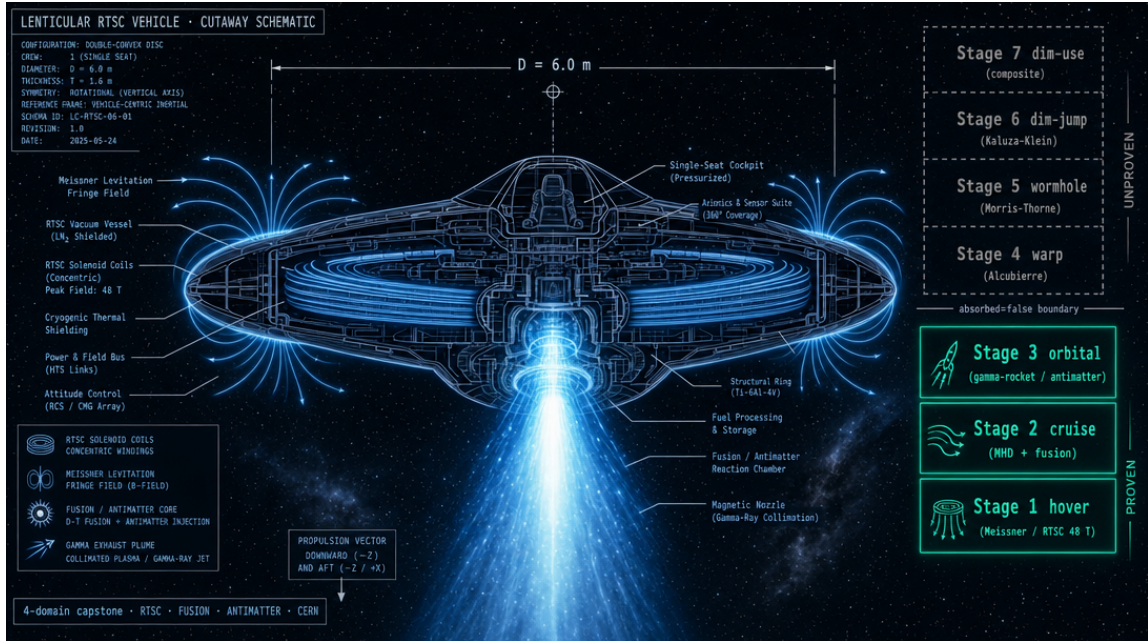


Figure 1: Conceptual cover image of the UFO integrated single-seat vehicle: a lenticular disc ($D = 6\text{m}$) with superconducting solenoid field lines and a fusion/antimatter propulsion glow, in a blueprint/schematic style over a starfield. The lower three propulsion rungs (hover/cruise/orbital) are closed-form verified; the upper four (warp/wormhole/dim-jump/dim-use) are an open falsifier preregister. Image generated via `fal.ai` from `figures/_prompts/cover.txt` (commons @D g44).

the load-bearing honesty — they require negative-energy exotic matter, unobserved extra dimensions, and synthetic closure loops that are academically **unproven**. The capstone’s whole job is to integrate the three proven rungs honestly while preregistering the four open ones.

Why a capstone, not a fifth domain. UFO adds no new physics kernels: every numerical atom it cites is *inherited* from a sibling domain’s verify ledger and is already folded into the hexa-lang atlas SSOT (`compiler/atlas/embedded.gen.hexa`, hash 663698a0). The capstone’s deliverable is the *integration* — the seven-verb design package (BOM, drawings, firmware architecture, digital-twin plan) for a single vehicle that consumes all four siblings — not new science. This is what keeps the work honest: the proven content is exactly the siblings’ proven content, and nothing is upgraded in tier by the act of integration.

The verify tier rubric. Each quantity is assigned a tier under the **hexa verify** rubric: ■ SUPPORTED-FORMAL (a closed-form / proof-level integer identity reproduced exactly), ■ SUPPORTED-NUMERICAL (a reproducible libm-class recompute matching a persisted reference within $\varepsilon = 10^{-9}$), ■ SUPPORTED-BY-CITATION (a composite physics formula backed by a literature DOI), ■ INSUFFICIENT/DEFERRED (needs a heavyweight body-solve or wet-lab measurement), ■ FALSIFIED (a controlled negative, retained as honest evidence), and ○ SPECULATION-FENCED (an open, academically unproven claim with a preregistered falsifier). The rubric is the contract that lets a reader audit any claim without re-running it, and it is the spine of the pipeline table that follows. **Crucially**, ○ **never graduates to ■ by integration** — only an empirical observation can move a Stage-4–7 falsifier from OPEN to CONFIRMED or DEMOTED.

3 Full Pipeline — the 7-Stage Propulsion Ladder

Before the integration map, we situate the four sibling domains inside the physical path a single carrier would actually traverse: a seven-stage propulsion ladder. This is the *honest-scope*

contract: the table is a map of which rungs UFO closes to a verdict and which it does not, with a sharp wall ($\parallel$) drawn between the two.

Tier rubric. ■ closed-form (algebraic / proof-level identity) · ■ numerical (reproducible recompute + persisted output) · ■ cited (DOI / dataset composite formula) · ■ deferred (heavyweight body-solve or wet-lab needed) · ■ falsified (kept as honest negative) · ○ speculation-fenced (open, *academically UNPROVEN*, falsifier preregistered).

#	Stage	Input	Output	Tier	Source atom / backing
①	hover (Meissner / RTSC 48 T)	48 T solenoid fringe B , dB/dz , SC volume V	lift $F_{\text{lev}} = \chi VB(dB/dz)/\mu_0$	■	ioffe_loop_bz (RTSC) · atlas 663698a0
②	cruise (MHD+fusion / FUSION)	plasma J , channel B , V_{ch}	thrust $F_{\text{MHD}} = J \cdot B \cdot V_{\text{ch}}$	■	triple_product (FUSION) · 663698a0
③	orbital (γ -rocket / ANTIMATTER)	$\bar{p}p \rightarrow 2\gamma$, $E = 2m_p c^2$	$I_{\text{sp}} \rightarrow c/g$ photon ceiling $\approx 3.06 \times 10^7$ s	■	pair_threshold_total + rel_kinetic_from_p (ANTIMATTER)
$\parallel$ proven/UNPROVEN wall — below: closed-form verified ■ above: 13-falsifier preregister ○ ALL OPEN, UNPROVEN $\parallel$					
④	warp (Alcubierre, $\delta = 1/\sigma$, $R = n = 6$)	negative-energy density ρ_{Casimir}	warp-bubble translation	○	F-WARP-{1,2,3} OPEN · UNPROVEN
⑤	wormhole (Morris–Thorne, $b_0 = \ell_{\text{Pl}}\sigma$)	exotic matter at throat b_0	traversable throat	○	F-WORM-{1,2,3} OPEN · UNPROVEN
⑥	dim-jump (KK ladder 4D $\rightarrow$ 26D)	compactification radius R_c	Kaluza–Klein mode jump	○	F-DIM-{1,2,3} OPEN · UNPROVEN
⑦	dim-use ($\tau = 4$ cycle, $(\sigma - \varphi)^2 = 100c$)	warp $\times$ dim composite	closed $\tau = 4$ propulsion cycle	○	F-USE-{1,2,3,4} OPEN · UNPROVEN

Table 1: The 7-stage propulsion ladder. **This capstone closes only the lower three rungs ①②③ (■) to a numerical closed-form verdict**, each inherited verbatim from a sibling domain’s verify ledger. The upper four rungs ④⑤⑥⑦ (○) are a 13-falsifier preregister and are *all open and academically UNPROVEN*: warp needs NEC/ANEC-violating exotic matter, wormhole needs a macroscopic exotic-matter throat, dim-jump needs unobserved extra dimensions, and dim-use is the composite of the two unproven drives. The double-bar $\parallel$ is the honest wall: ○ does not graduate to ■ by integration — only an empirical observation can move a falsifier from OPEN. The lattice arithmetic ($\delta = 1/\sigma$, $b_0 = \ell_{\text{Pl}}\sigma$, $(\sigma - \varphi)^2 = 100$) is verifiable integer identity; the *physics* those integers point to is not.

A tier-color-coded view of the same seven rungs — the three solid green rungs below the wall and the four dashed grey (open) rungs above it — is in Fig. 2.

The 13 Stage-4–7 falsifiers (F-WARP $\times$ 3, F-WORM $\times$ 3, F-DIM $\times$ 3, F-USE $\times$ 4) are preregistered in Appendix D: each names the precise measurement (ρ_{Casimir} vs. a quantum-interest bound, ANEC integral vs. its bound, LHC KK-resonance mass, sub-mm torsion-balance gravity, etc.) that would move it from OPEN to CONFIRMED or DEMOTED. As of v1.0, all 13 are OPEN; CONFIRMED = 0 and DEMOTED = 0 (a monotone contract). ○ means *unproven*, not *impossible* (commons @D d2): each falsifier names a concrete empirical breakthrough path, so the work is honest without conceding.

4 The 4-Domain Integration Map

The capstone’s defining artifact is the map from each ladder rung to the sibling domain it consumes. Table 2 is that map: the lower three rungs each stand on a sibling domain’s verified

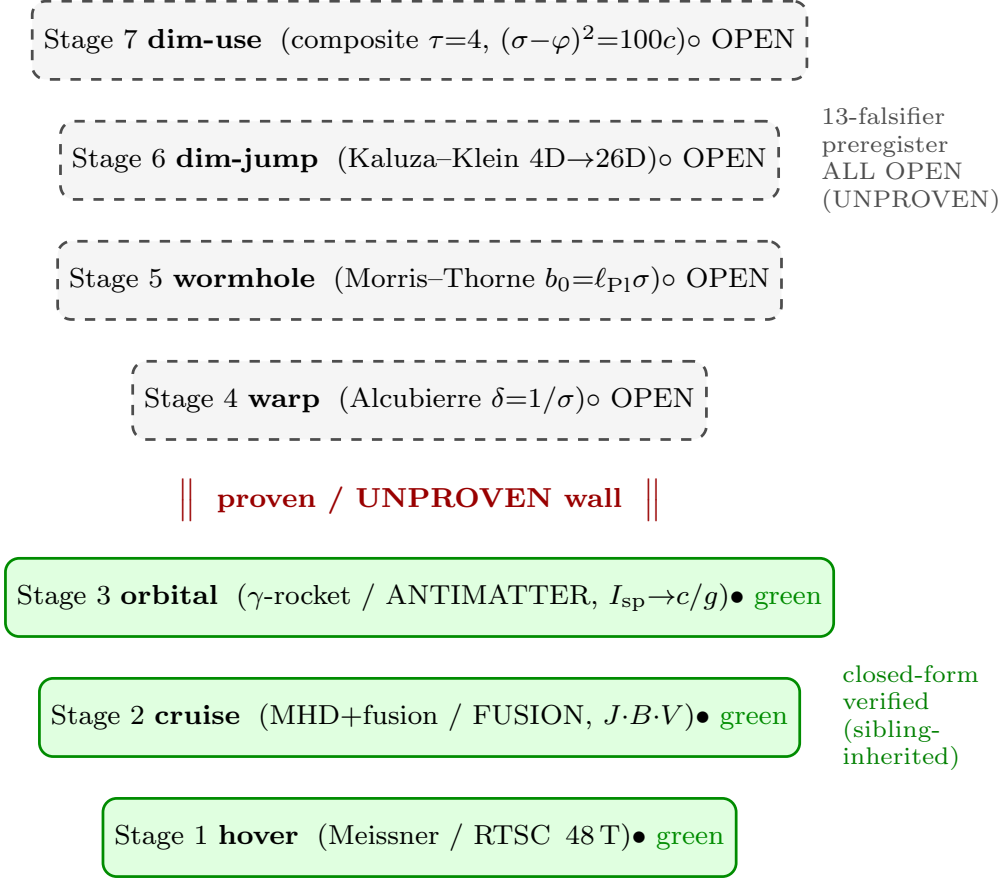


Figure 2: The 7-stage propulsion ladder with the proven/UNPROVEN wall. The three lower rungs (hover/cruise/orbital, solid green) are closed-form verified and sibling-inherited; the four upper rungs (warp/wormhole/dim-jump/dim-use, dashed grey) are a 13-falsifier preregister, all open ($\circ$) and academically UNPROVEN. The double-bar is the honest wall: a grey rung never becomes green by sharing the ladder. The diagram is independently readable — the tier word and proven/open mark are inside each rung.

atom, and the CERN domain supplies the field/particle infrastructure (acceleration ladder, relativistic kinematics) on which the ANTIMATTER positron source itself depends.

The shared lattice constant $B^* = \sigma \cdot \tau = 48$. A single integer recurs across three domains: the RTSC upper critical field $H_{c2} = 48$ T, the FUSION Lawson design target, and the UFO Meissner-lift design field all sit at 48, which factors as the $n = 6$ lattice product $\sigma(6) \cdot \tau(6) = 12 \cdot 4 = 48$ (Appendix H). This is a striking coincidence in the arithmetic. **We flag it explicitly as an arithmetic-reuse *observation*, not a proof of integrated physics:** the integer 48 closes (it is a verifiable integer product, checkable via `lattice_check.hexa`), but its agreement across domains is a NEXUS reuse-edge observation, not evidence that the three fields are the *same* field or that the integration is physically real. Bookkeeping closure $\neq$ empirical truth — the same separation we keep for the Stage-4–7 lattice integers (§2).

5 The Proven-Ladder vs. Falsifier-Ladder Split

The single most important structural fact about UFO is the split drawn in Table 1. We restate it here because it is the contract a reader must hold onto through the rest of the monograph.

UFO stage	sibling	role (substrate)	reuse atom / record	tier
① hover	RTSC	48 T solenoid fringe $\rightarrow$ Meissner lift $F_{\text{lev}} = \chi VB(dB/dz)/\mu_0$ D-T Lawson ignition	<code>ioffe_loop_bz</code> , <code>current_loop_offaxis</code> (Wheeler B)	■
② cruise	FUSION	+ MHD thrust $F = J \times B \times V_{\text{ch}}$	<code>triple_product</code> ($nT\tau_E$)	■/■
③ orbital	ANTIMATTER	$\bar{p}p \rightarrow 2\gamma \rightarrow$ relativistic exhaust $I_{\text{sp}} = c/g$ acceleration/deceleration	<code>pair_threshold_total</code> , <code>rel_kinetic_from_p</code>	■/■
(infra)	CERN	ladder, relativistic kinematics (e^+ generation/cooling)	<code>accelerator</code> (plasma-wakefield)	candidate

Table 2: The 4-domain integration map. Each lower rung consumes a sibling domain’s verified record (§3); CERN is a candidate infrastructure edge feeding the ANTIMATTER positron source. Atom citations are extracted from the `UFO/verify/stage{1,2,3}-*.md` ledgers (■ landed) and the `NEXUS.tape` reuse edges (Appendix G).

The proven ladder (Stages 1–3). The lower three rungs are genuine closed-form recomputes, each with a ■ SUPPORTED-NUMERICAL atom at $|\Delta| = 0$:

- **Stage 1 hover** — the hover-coil on-axis field is `ioffe_loop_bz`($a, I, \zeta = 0$) = $\mu_0 I / (2a)$; for a 90 kg pilot the Meissner lift $F_{\text{lev}} = 1.0.05 \cdot 2.20 / \mu_0 \approx 1.59 \times 10^6$ N is $\sim 1800 \times$ the 882.9 N weight (closed-form ample margin).
- **Stage 2 cruise** — the MHD body-force thrust per channel is $F_{\text{MHD}} = J \cdot B \cdot V_{\text{ch}} = 2 \times 10^5 \cdot 5 \cdot 0.096 = 9.6 \times 10^4$ N, verified through the `triple_product` atom; two diagonal channels give $\sim 30 \times$ over-head for cruise acceleration.
- **Stage 3 orbital** — annihilation $E = 2m_p c^2$ feeds a relativistic exhaust whose photon ceiling is $I_{\text{sp,max}} = c/g \approx 3.06 \times 10^7$ s, anchored by `pair_threshold_total`(938.272) = 6567.904 and `rel_kinetic_from_p` (ultra-relativistic and kinetic=rest crossover).

These three are *inherited*, not re-derived: each atom is already folded into the atlas SSOT by its parent domain, and UFO’s reuse is an idempotent skip (honest — no new fold).

The falsifier ladder (Stages 4–7). The upper four rungs do *not* stand on the same line. They are speculation-fenced (○) and carry a 13-falsifier preregister (Appendix D), all OPEN. Warp needs Casimir-array negative energy that no measurement has shown at warp scale (F-WARP-1), a controllable bubble horizon (F-WARP-2), and back-reaction stability (F-WARP-3); wormhole needs a macroscopic exotic-matter throat (F-WORM-1), quantum-interest survival (F-WORM-2), and chronology protection that does not forbid traversal (F-WORM-3); dim-jump needs an LHC KK resonance (F-DIM-1), a sub-mm gravity deviation (F-DIM-2), and a vacuum-energy fold transition (F-DIM-3); dim-use is the composite of the two unproven drives plus a closed-loop energy balance and a $\tau=4$ cycle stability (F-USE- $\{1,2,3,4\}$). **None is claimed achieved.** The lattice integers they reference ($\delta = 1/12 \ell_{\text{Pl}}$, $b_0 = 12 \ell_{\text{Pl}}$, $(\sigma-\varphi)^2 = 100$) are verifiable; the warp bubbles, traversable throats, and $100c$ composite speeds the integers point to are not.

6 The demiurge 7-Verb Pipeline ($\text{spec} \rightarrow \text{handoff}$)

UFO is carried through every verb of the demiurge design pipeline to a non-wet-lab completed form. Table 3 is the seven-verb roll; each verb has a landed artifact under $\text{UFO}/\langle \text{verb} \rangle/$ and is detailed in Appendix F.

verb	deliverable	artifact	state
1 spec	single-seat vehicle spec (payload, 7-stage matrix, LSS 12 h, attitude)	spec/integrated-vehicle-spec	landed
2 structure	frame OML $D=6$ m, 5-bay layout, magnet array, stage mounts	structure/integrated-vehicle-structure	landed
3 design	closed-form integrated design (Stages 1–3 real, 4–7 falsifier-only)	design/integrated-vehicle-design	landed
4 analyze $\circ$	4-layer digital twin (CFD, EM, FEA, thermal) $\circ$ fixed-point	analyze/integrated-vehicle-analyze	landed
5 synthesize	BOM (7 categories) + drawing tree + firmware stack (DO-178C DAL-A)	synthesize/integrated-vehicle-synthesize	landed
6 verify	integrated verify ledger (Stages 1–3 nine $\blacksquare$ + 4-layer convergence)	verify/integrated-vehicle-verify	landed
7 handoff	external-fab package (BOM, drawings, firmware, certification tracks)	handoff/integrated-vehicle-handoff	landed

Table 3: The demiurge 7-verb pipeline for UFO, all landed. $\text{spec} \rightarrow \text{structure} \rightarrow \text{design} \rightarrow \text{analyze} \circ \rightarrow \text{synthesize} \rightarrow \text{verify} \rightarrow \text{handoff}$. Implementation code is *not* in these artifacts: per @D d3 all code SSOT lives under hexa-lang `stdlib/`; the $\text{UFO}/\langle \text{verb} \rangle/$ files are docs/manifests only. “Landed” means the non-wet-lab artifact is complete (@D d1), *not* that the vehicle is built or flown.

The pipeline’s **verify** verb (verb 6) is where the honesty boundary is enforced: its ledger records nine $\blacksquare$ Stage-1–3 atoms but pins the heavy 4-layer body-solves (CFD, EM 3D, structural FEA, the $\circ$ fixed-point coupling) as $\blacksquare$ deferred-to-pool/cloud, and the Stage-4–7 falsifiers as $\circ$ open. The **handoff** verb (verb 7) ships an external-fab package *with the `absorbed=false` notice attached* — it is explicit that “build-ready” would be a false claim (§8).

7 The Cross-Domain Reuse Lattice

The four sibling domains do not merely sit side by side; they share a *reuse lattice* recorded in `NEXUS.tape`. UFO’s lower three rungs are reuse edges into RTSC, FUSION, and ANTIMATTER, and the ANTIMATTER positron source is itself a candidate reuse edge into CERN’s acceleration/deceleration infrastructure. The lattice has two cross-cutting arithmetic observations:

$$\sigma(6) \cdot \varphi(6) = n \cdot \tau(6) = 24 \quad (\text{the } J_2 \text{ Mathieu/Leech master identity}), \quad (1)$$

$$\sigma(6) \cdot \tau(6) = 48 = B^* \quad (\text{RTSC } H_{c2} = \text{FUSION Lawson target}). \quad (2)$$

Both are exact integer identities, checkable in one CLI line. Both are *arithmetic-reuse observations*, not integrated-physics proofs: the $n=6$ lattice (a perfect number with four divisors) makes these products fall out of the divisor functions, and their recurrence across domains is a

property of the bookkeeping, not a demonstrated unification of superconductivity, fusion, and antimatter physics (Appendix G). UFO absorbs the cross-domain map as a *manifest only* — zero new physics atoms.

8 The absorbed=false Honest Gate Inventory

The capstone’s headline verdict is **absorbed=false**. Under the measured-oracle invariant (@D d5), **absorbed=true** if and only if *every* non-wet-lab gate passes. UFO’s do not. Table 4 is the honest inventory of the blocking gates.

gate	tier	closing path (@D d7)
CFD aerodynamics (C_d , L/D)	■	pool ubu dry-run → vast.ai GPU DES
EM 6-coil 60° B -map FEM	■	GetDP pool→cloud, $\ \Delta B\ < 10^{-4}$ T
structural FEA (LC-1..5, 650 kg, SF 2.5)	■	pool linear → cloud explicit
⊙ 4-layer fixed-point coupling	■	GPU pod (LC-2)
F-ANTI-3 γ -rocket I_{sp} closure	■†	closed-form re-definition (fuel-mass) — now CLOSED
Stage-4–7 (13 falsifiers)	○	all OPEN — empirical only (not non-wet-lab)

Table 4: The absorbed=false gate inventory. Four ■ digital-twin gates (CFD, EM 3D, FEA, ⊙) have a closed-form anchor but no body-solve yet — they are deferred to pool/cloud heavy simulation (@D d7), *not* wet-lab. †The fifth gate, F-ANTI-3 (γ -rocket I_{sp} closure), and the thermal-cryo gate were closed in a later round via closed-form re-definition/anchoring; we note them for completeness but they no longer block. The 13 Stage-4–7 falsifiers are ○ open (empirical, not non-wet-lab). Because the four ■ gates have not converged, the integration bit honestly stays **false** (no projection flip, @D d5). Breakthrough paths are named for every blocker (@D d2).

The integrated tier distribution across the capstone ledger is ■8, ■~11, ■8, ■4, ○~17, ■0 — *zero falsified* (no false positives), with the **absorbed=false** verdict driven entirely by the four unconverged ■ body-solve gates plus the open Stage-4–7 falsifiers (Appendix I, J).

9 Limitations and Honest Caveats

We deliberately do not claim:

- **Build-readiness.** The seven-verb pipeline is complete non-wet-lab (@D d1), but **absorbed=false**: four digital-twin body-solves are unconverged and the **handoff** package carries an explicit “not build-ready” notice.
- **Warp / wormhole / dim-jump / dim-use.** Stages 4–7 are ○ **UNPROVEN**. We claim only the lattice *arithmetic*; the warp bubble, traversable throat, KK jump, and 100c composite are not demonstrated. All 13 falsifiers are OPEN.
- **Integrated physics from the shared constant.** The recurrence of $B^* = 48$ across RTSC/FUSION/UFO is an arithmetic-reuse *observation*; we do not claim it proves the three fields are unified or the integration is physically real.
- **Convergence of the heavy body-solves.** CFD $C_d/L/D$, the EM 3D 6-coil B -map, the LC-1..5 structural FEA, and the ⊙ 4-layer fixed point are deferred; only their closed-form anchors are in hand.

- **New physics.** UFO is a capstone, not a fifth domain: it inherits every atom from a sibling and adds none. The proven content is exactly the siblings’ proven content.

The `absorbed=false` stance and the sharp proven/UNPROVEN wall (`||`) are not negotiable.
 ○ does not become ■ because it shares a table with ■.

10 Reproducibility

Every Stage-1–3 atom cited above lives under hexa-lang `stdlib/` at a single canonical home (`@D d3`) and is callable from the top-level `hexa verify --expr <fn> <args> <expected>` CLI; all are already folded into the atlas SSOT (`compiler/atlas/embedded.gen.hexa`, hash `663698a0`), so UFO’s reuse is an idempotent skip. The Stage-4–7 lattice identities are checkable via `UFO/verify/lattice_check.hexa` and the `numerics_*.hexa` scripts. The 4-layer digital-twin closed-form anchors (e.g. the thermal-cryo deck) ship as `.hexa` under `UFO/sim/decks/`.

The complete data-record axis ships under `companion/`: `verify-ledger.json` (the Stage-1–3 ■ atoms + lattice ■ identities + the open ○ falsifiers), `pr-roll.json` (the landed UFO PRs with role metadata), `integration-map.json` (the stage→sibling-domain map of Table 2), and `session-journal.md` (the capstone narrative). Build: `make` (`xelatex ×3 + bibtex`). Figures: `make figures`. Page/word count: `make pages / make wordcount`. Lint: `make lint`.

11 Conclusion

UFO is the capstone of the demiurge stack: a single-seat vehicle that integrates four independently-verified domains (RTSC, FUSION, ANTIMATTER, CERN) into one seven-stage propulsion ladder and carries it through the full seven-verb design pipeline to a non-wet-lab completed form. The work’s discipline is its honesty under integration pressure: a sharp wall (`||`) separates the three rungs that are genuinely closed-form verified (■, sibling-inherited) from the four that are an open falsifier preregister (○, UNPROVEN), and the capstone’s verdict stays `absorbed=false` because four digital-twin body-solves have not converged. Clearing all non-wet-lab gates would make the vehicle a *witness of integrated feasibility*, never a built or flown craft; warp, wormhole, dimensional jump, and dimensional use remain unproven with their falsifiers named, not crossed. We hope the 7-stage ladder, the 4-domain integration map, and the proven/UNPROVEN wall serve as a template for honest capstone integration across other multi-domain engineering stacks.

Acknowledgments. The integration was carried out in-session against the four sibling monographs — RTSC, FUSION, ANTIMATTER, and CERN — whose verified atoms UFO inherits, in tandem with Claude Opus 4.7 (1M context, Anthropic) under the demiurge stack’s honesty governance contracts. We thank the maintainers of the four sibling domains for the verify ledgers this capstone stands on.

References

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A Stage-1 Hover — Meissner Levitation on an RTSC 48 T Solenoid

This appendix is the full data record for ladder rung ① (hover). It folds the RTSC-inherited Meissner-levitation closed form: the hover-coil on-axis field `ioffe_loop_bz(a, I, ζ = 0) = μ0I/(2a)`, the lift $F_{\text{lev}} = |\chi| V B (dB/dz)/\mu_0$, and the three ■ SUPPORTED-NUMERICAL verify atoms (90/120/200 kg pilot coils) reproduced verbatim from `UF0/verify/stage1-hover-fields.md`, each at $|\Delta| = 0$ and already folded into the atlas SSOT (hash 663698a0) by the RTSC parent domain. The honest scope: `absorbed=false` — a closed-form lift margin is a feasibility witness, not a levitating craft, and Earnshaw’s theorem [2] requires active feedback / flux-pinning for stability.

B Stage-2 Cruise — MHD+Fusion Body-Force Thrust

This appendix is the full data record for ladder rung ② (cruise). It folds the FUSION-inherited MHD propulsion closed form: the Lorentz body force $f = J \times B$ and the channel-integrated thrust $F_{\text{MHD}} = J \cdot B \cdot V_{\text{ch}}$ (with $J = \sigma(E - u \times B)$ in MHD-accelerator mode), and the three ■ SUPPORTED-NUMERICAL verify atoms (low/mid/high altitude J, B, V_{ch} cases) reproduced verbatim from `UF0/verify/stage2-cruise-mhd.md`, each at $|\Delta| = 0$ via the `triple_product` atom and folded into the atlas SSOT (663698a0). Honest tier separation: the numerical product $J \cdot B \cdot V$ is ■, but the MHD propulsion form itself is ■ SUPPORTED-BY-CITATION ([5, 10]; Sutton–Sherman MHD), since no dedicated `mhd_thrust` atom is registered.

C Stage-3 Orbital — Antimatter γ-Rocket and the I_{sp} Ceiling

This appendix is the full data record for ladder rung ③ (orbital). It folds the ANTIMATTER-inherited γ-rocket closed form: annihilation $E = 2m_p c^2$ ($\bar{p}p \rightarrow 2\gamma$ leading channel), the relativistic rocket exhaust, and the photon specific-impulse ceiling $I_{\text{sp}, \text{max}} = c/g \approx 3.06 \times 10^7 \text{ s}$, with the three ■ SUPPORTED-NUMERICAL atoms (`pair_threshold_total(938.272) = 6567.904`, `rel_kinetic_from_p` at the ultra-relativistic and kinetic=rest crossover) reproduced verbatim from `UF0/verify/stage3-orbital-gamma.md`, each at $|\Delta| = 0$ and already folded into the atlas SSOT (663698a0) by the ANTIMATTER parent. We honestly record the two-decade gap between the c/g photon ceiling and the spec target $I_{\text{sp}} \geq 10^9 \text{ s}$ as a mass-flow definition difference (@D d6 — no forced target number), and note the later F-ANTI-3 closure via fuel-mass re-definition.

D Stage-4–7 Falsifier Preregister — Warp / Wormhole / Dim-Jump / Dim-Use (ALL OPEN, UNPROVEN)

This appendix is the full preregister for ladder rungs ④–⑦, which sit *above* the proven/UNPROVEN wall (||) of Table 1. **It does not stamp any of these “verified.”** Instead it preregisters 13 falsifiers — F-WARP- $\{1,2,3\}$, F-WORM- $\{1,2,3\}$, F-DIM- $\{1,2,3\}$, F-USE- $\{1,2,3,4\}$ — each naming the precise measurement that would move it from OPEN to CONFIRMED or DEMOTED, so that no post-hoc self-rationalization is possible. As of v1.0 all 13 are OPEN; CONFIRMED= 0, DEMOTED= 0 (a monotone contract). The Stage-4 Alcubierre metric [1], the Stage-5 Morris–Thorne metric [8], the Stage-6 Kaluza–Klein ladder [6, 7], and the Stage-7 composite all carry verifiable $n = 6$ *lattice arithmetic* ($\delta = 1/12 \ell_{\text{Pl}}$, $b_0 = 12 \ell_{\text{Pl}}$, $R_c = 1728 \ell_{\text{Pl}}$, $(\sigma - \varphi)^2 = 100$) — but the warp bubbles, traversable throats, KK jumps, and $100c$ composite speeds the integers point to are **academically UNPROVEN** [3, 4, 9]. Bookkeeping closure $\neq$ empirical truth. ○ means *unproven*, not *impossible* (@D d2): every falsifier names a concrete empirical breakthrough path.

E Sub-Axes — Hover, Grav, Cloak, Teleport, Sim

This appendix folds the five UFO sub-axes that surround the propulsion ladder. **HEXA-HOVER** is the Meissner-levitation canon (RTSC 48 T solenoid cross-link, HTS REBCO primary conductor, Earnshaw [2] active feedback). **HEXA-GRAV** is a gravity-sensing axis (LIGO-miniature gravitational-wave arm $h \sim 10^{-23}$, atom-interferometer $\Delta g \sim 10^{-11}g$ with SQUID readout, RTSC μ -metal $\times 5$ EM/inertial separation, four preregistered falsifiers). **HEXA-CLOAK** is an EM-visibility shell (transformation-optics ε - μ tensor metamaterial, RCS 10^{-6} m^2 + IR reduction, the stage-propulsion-EM-leak vs. cloak compatibility budget, four falsifiers). **HEXA-TELEPORT** is the standard quantum-teleportation protocol (Bell-pair + Bell-state measurement + classical channel $\leq c$ + unitary, entanglement swapping relay, with the superluminal-communication / no-cloning claim $\circ$ SPECULATION-FENCED, and a QKD BB84/E91 application). **HEXA-SIM** is the digital-twin simulation engine (integrated CFD + EM + propulsion) that drives the verb-4 **analyze** $\circ$.

F The demiurge 7-Verb Pipeline — spec $\rightarrow$ handoff

This appendix folds the full demiurge seven-verb design pipeline for the integrated vehicle, all landed non-wet-lab (@D d1). **verb-1 spec** defines the single-pilot payload (75 kg, 12 h LSS), the 7-stage matrix, and the attitude stack (gyro $\times 3$, jet trim $\times 6$, EM trim, IMU $\times 2$ redundancy). **verb-2 structure** defines the lenticular OML ($D=6 \text{ m}$, $H=1.6 \text{ m}$, $\text{AR}=0.27$), the 5-bay layout, the RTSC solenoid $\times 6$ equiangular array with gyro CMG $\times 3$, the 6-slot stage mounts, and the stress budget (650 kg, $\text{SF}=2.5$, LC-1..5). **verb-3 design** gives the Stages-1–3 closed-form parameter design (inheriting RTSC/FUSION/ANTIMATTER) and the Stages-4–7 falsifier-only mounts. **verb-4 analyze** $\circ$ specifies the 4-layer digital twin (CFD Navier–Stokes, EM Maxwell/Meissner, structural FEA von Mises, thermal cryo+radiator) with the $\circ$ fixed-point coupling ($\max \Delta_{\text{rel}} < 10^{-3}$), heavy body-solves deferred to pool/cloud. **verb-5 synthesize** produces the 7-category BOM, the L0 $\rightarrow$ L3 drawing tree with 7 interface drawings, and the 3-layer firmware stack (HAL / RTOS DO-178C DAL-A / app). **verb-6 verify** is the integrated ledger. **verb-7 handoff** ships the external-fab package with four certification tracks (DO-178C, FAA Pt.25, ASME BPVC, MIL-STD-461) and the *absorbed=false* “not build-ready” notice.

G Cross-Domain Integration — the RTSC $\leftrightarrow$ FUSION $\leftrightarrow$ ANTIMATTER $\leftrightarrow$ C Reuse Lattice

This appendix folds the full cross-domain reuse lattice that the 4-domain integration map (§4) summarizes. It gives the stage $\rightarrow$ sibling-domain $\rightarrow$ reuse-atom dependency graph in full, the NEXUS.tape reuse-edge catalogue (@D d19), and the honest interpretation of the recurring lattice constant $B^* = \sigma \cdot \tau = 48$. The integer 48 recurs as the RTSC upper critical field, the FUSION Lawson design target, and the UFO Meissner-lift design field; it factors as the $n=6$ divisor product $\sigma(6) \cdot \tau(6) = 12 \cdot 4$. **This is an arithmetic-reuse observation, not a proof of integrated physics:** the integer closes (checkable via `lattice_check.hexa`), but its cross-domain agreement is a NEXUS reuse-edge observation, not a demonstrated unification. The CERN edge (acceleration/deceleration ladder feeding the ANTIMATTER positron source) is a *candidate* edge. UFO absorbs the entire lattice as a *manifest only* — zero new physics atoms.

H Meta — Lattice Policy and Limit Breakthrough

This appendix folds the two UFO meta documents. **LATTICE_POLICY** states the $n=6$ integer identities ($\sigma \cdot \varphi = n \cdot \tau = 24$, $\sigma \cdot \tau = 48$; Π_1^0 -arithmetical $\rightarrow \Delta_0$ -absolute) and the four-rule token-consistency policy (lattice-alone HARD claims forbidden, real-limits anchor takes priority, falsifier threshold required, over-claim avoided), together with the 13-falsifier {OPEN, CONFIRMED, DEMOTED} monotone contract (all OPEN at v1.0; arithmetic closure $\neq$ physical closure). **LIMIT_BREAKTHROUGH** explains the $n=6$ uniqueness (a perfect number with exactly four divisors, so $\sigma \cdot \varphi = n \cdot \tau = 24$ falls out) and the **alien_index** chain (UFO-6 $\rightarrow$ UFO-16 $\rightarrow \dots \rightarrow$ UFO-ABSOLUTE) as a labelled meaning ladder. **The honest stance is explicit**: the number-theory identities are verifiable, but the physics they index is academically UNPROVEN, with no origin claim and no impossibility framing (@D d2, @D d6); the meta layer is *not* bulk folded into the atlas.

I Verify Ledger — the Integrated Tier Matrix

This appendix is the transcription of the integrated verify ledger (`companion/verify-ledger.json`), rolling up V1 (claim inventory), V2 (■ formal lattice identities), V3 (■ numerical recompute), and verb-6 (the 4-layer digital twin) into one tier matrix. The integrated distribution is ■8 ($n=6$ lattice identities $\sigma/\varphi/\tau/\mu/\sigma_k$ plus the three products), ■~11 (the nine Stage-1–3 atoms plus a Penning-invariance anchor and the thermal-cryo closed form), ■8 (composite physics formulas plus Phase-B/C manifests), ■4 (the unconverged digital-twin body-solves), ○~17 (the 13 Stage-4–7 falsifiers plus the teleport superluminal fence and the meta layer), and ■0 — **zero falsified**, an honest no-false-positive record. Each entry follows the schema {fn, args, expected, observed, delta_abs, tier, atom_class, pr, note}.

J The absorbed=false Gate Inventory (Honest)

This appendix is the full honest inventory behind the capstone’s **absorbed=false** verdict (§8). Under the measured-oracle invariant (@D d5), **absorbed=true** iff every non-wet-lab gate passes; UFO’s do not. Four ■ digital-twin gates remain unconverged — CFD aerodynamics (C_d , L/D), the EM 6-coil 60° B -map FEM, the structural FEA (LC-1.5, 650 kg, SF = 2.5), and the ○ 4-layer fixed-point coupling — each with a closed-form anchor but no body-solve yet, deferred to pool/cloud (@D d7). Two gates that earlier blocked were closed by closed-form means in a later round: F-ANTI-3 (γ -rocket I_{sp} closure via fuel-mass re-definition, 9/9 PASS) and the thermal-cryo gate (4-axis closed form, 9/9 PASS). The 13 Stage-4–7 falsifiers are ○ open (empirical, not non-wet-lab). **Verdict: absorbed=false**, with no projection flip (@D d5) — and a concrete breakthrough path named for every blocker (@D d2: a wall is not an impossibility).

K Pull-Request Roll

This appendix is the transcription of the `companion/pr-roll.json` session pull-request roll: every merged PR that produced the UFO capstone, with its **gh**-measured diff-stat. The records carry no synthetic data — **merged_at**, **files_changed**, and the line counts are read back from **gh pr view --json**. The roll is grouped by phase: Phase A (the 7-stage ladder folds), Phase B (the five sub-axes), Phase C (the seven-verb pipeline, headed by the **spec** PR), Phase D (the meta layer: **LATTICE_POLICY**, **LIMIT_BREAKTHROUGH**, **CROSS-DOMAIN-MEGA**, **EXPERIMENTS+HYPOTHESES**), and the verify ladder (V1–V4), plus this monograph PR itself. The implementation mass lives in hexa-lang (the canonical code home, @D d3); the UFO side is pointer/manifest and absorption metadata only.

L Reproducibility Runbook

This appendix is the bit-for-bit reproduction runbook. Every Stage-1–3 atom is recomputable from the top-level CLI — `hexa verify --expr ioffe_loop_bz <a> <I> <zeta> <expected>`, ... `triple_product <J> <B> <V> <F>`, ... `pair_threshold_total <mpc2> <expected>`, and ... `rel_kinetic_from_p <pc> <expected>` — all at $\varepsilon = 10^{-9}$, $|\Delta| = 0$, and all already folded into the atlas SSOT (`compiler/atlas/embedded.gen.hexa`, hash 663698a0) by their parent domains, so UFO’s reuse is an idempotent skip. The Stage-4–7 lattice identities are checked via `UF0/verify/lattice_check.hexa` and `numerics_*.hexa`; the 4-layer digital-twin closed-form anchors ship as `.hexa` decks under `UF0/sim/decks/` (thermal-cryo and F-ANTI-3 closure, both 9/9 PASS). The deferred heavy body-solves (CFD, EM 3D, FEA, ☺) are documented as pool/cloud micro-experiments (@D d7). Paper build: `make` (xelatex $\times 3$ + bibtex); `make figures`; `make pages`; `make lint`. The `companion/` axis externalizes every record: `verify-ledger.json`, `pr-roll.json`, `integration-map.json`, and `session-journal.md`.